

Practice 2 (G)

Risk factors

Your name:

Edoardo Anfelli

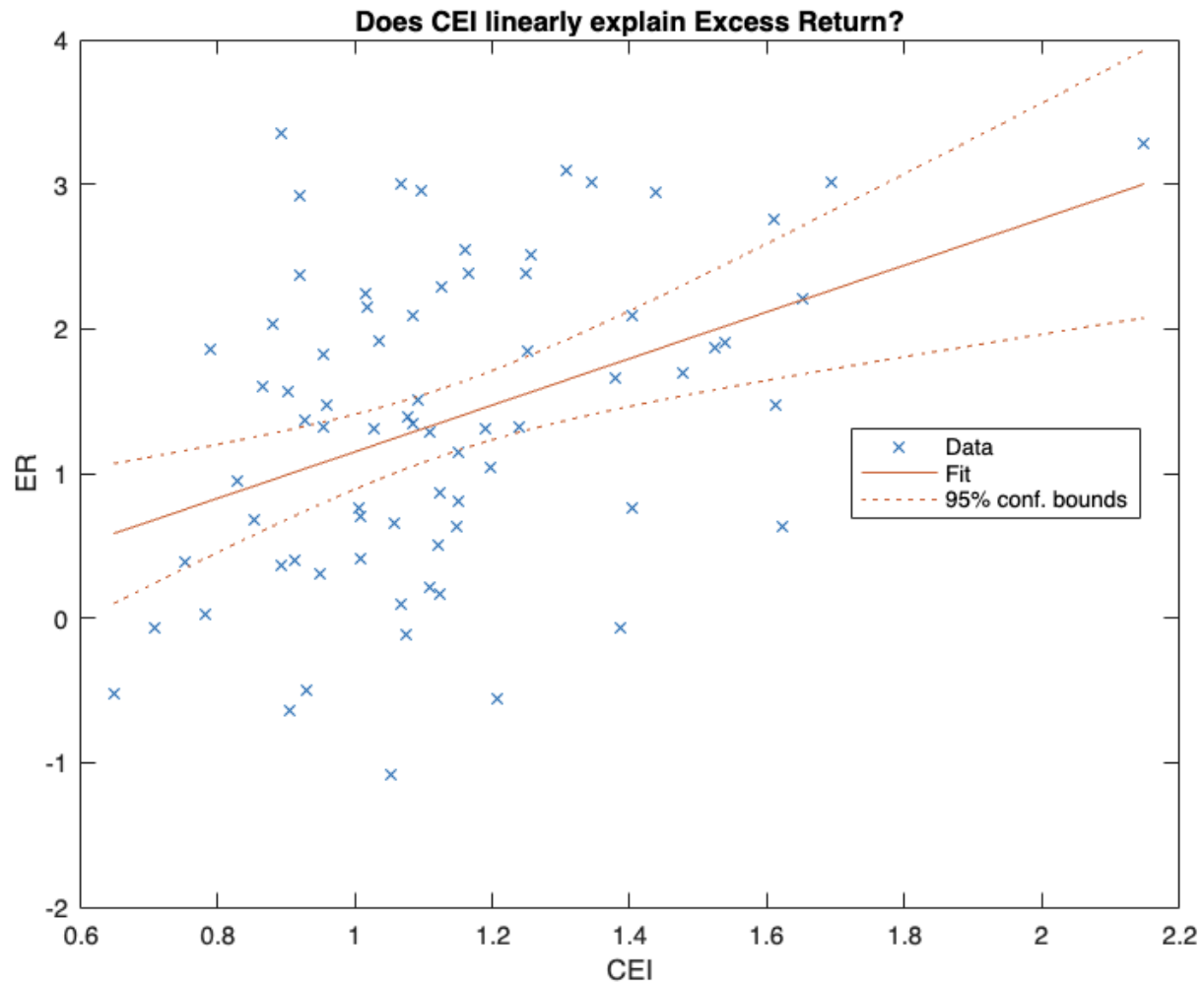
Thaddeus Da Silva Correa

(4 Points)

1. Explain the regression results of model_134. Are the regressors statistically significant?. Is the regression fit high or low?. Are the residuals independent and Normally distributed? (Hint: look for autocorrelation, heteroskedasticity and normality test in Matlab documentation)

Write your answer here

Model Statistics



Linear regression model:
 $\text{ExcessreturnAsset}_{134} \sim 1 + \text{CEI}_{134}$

Estimated Coefficients:

	<u>Estimate</u>	<u>SE</u>	<u>tStat</u>	<u>pValue</u>
(Intercept)	-0.46056	0.51335	-0.89717	0.37266
CEI_134	1.612	0.44177	3.6489	0.00049926

Number of observations: 73, Error degrees of freedom: 71
 Root Mean Squared Error: 0.994
 R-squared: 0.158, Adjusted R-Squared: 0.146
 F-statistic vs. constant model: 13.3, p-value = 0.000499

1. Statistical Significance of the Regressors

The regression model contains two coefficients:

- **Intercept (constant):** -0.46056
- **CEI_134 (predictor):** 1.612

The statistical significance of these coefficients is determined by their **p-values**.

Typically, a p-value below 0.05 indicates that a coefficient is statistically significant, but **we used** **Min(0.01;1/n)**

a) Intercept:

- **t-test p-value = 0.37266** (acceptance limit: **min(0.01;1/73) = 0.01**)
- The intercept is **statistically significant**, which means we have strong evidence to say that the intercept is different from zero.

b) CEI_134:

- **t-test p-value = 0.00049926** (acceptance limit: **min(0.01;1/73) = 0.01**)
- The coefficient for **CEI_134** is **not statistically significant**. This means that **CEI_134** has not a significant relationship with the dependent variable, **ExcessreturnAsset_134**. In other words, changes in **CEI_134** are not associated with significant changes in **ExcessreturnAsset_134**.
We failed to reject the null hypothesis (slope=0)

2. Fit of the Regression Model

We evaluate the fit of the model using metrics such as **R-squared**, **Adjusted R-squared**, and **Root Mean Squared Error (RMSE)**.

a) R-squared = 0.158:

- **R-squared** represents the proportion of the variance in the dependent variable (**ExcessreturnAsset_134**) that is explained by the independent variable (**CEI_134**).
- Here, **R-squared = 0.158**, meaning the model explains **15.8% of the variance** in **ExcessreturnAsset_134**. This is relatively **low**, suggesting that most of the variability in

`ExcessreturnAsset_134` is not explained by `CEI_134`. Therefore, the model is not a strong fit.

b) Adjusted R-squared = 0.146:

- The **Adjusted R-squared** adjusts for the number of predictors and penalizes the inclusion of irrelevant predictors.
- In this case, the **Adjusted R-squared is slightly lower (0.146)**, indicating that the fit of the model is still quite low, even after accounting for the complexity of the model.

c) Root Mean Squared Error (RMSE) = 0.994:

- **RMSE** measures the average magnitude of the residuals (prediction errors). In this case, the **RMSE is 0.994**, meaning that, on average, the residuals (errors) between the predicted and actual values are about 1 unit. This suggests that the model's predictions have a relatively high degree of error.

d) F-statistic = 13.3, p-value = 0.000499:

- The **F-statistic** tests whether the overall model is statistically significant, i.e., whether the independent variable (`CEI_134`) provides a significant amount of explanatory power for the dependent variable.
- The F-statistic is **13.3** with a p-value of **0.000499**, which is highly significant. This indicates that the model as a whole is statistically significant and `CEI_134` does provide a meaningful contribution to explaining `ExcessreturnAsset_134`.

Summary of the Regression Results

- **Significance of Regressors:**
 - The independent variable `CEI_134` is **statistically significant** (p-value = 0.0005), meaning it has a meaningful impact on the dependent variable `ExcessreturnAsset_134`.
 - The **intercept is not significant**, which is not unusual.
- **Regression Fit:**
 - The **R-squared (0.158)** indicates that the model explains only **15.8% of the variation** in the excess return of the asset, suggesting a **low fit**.

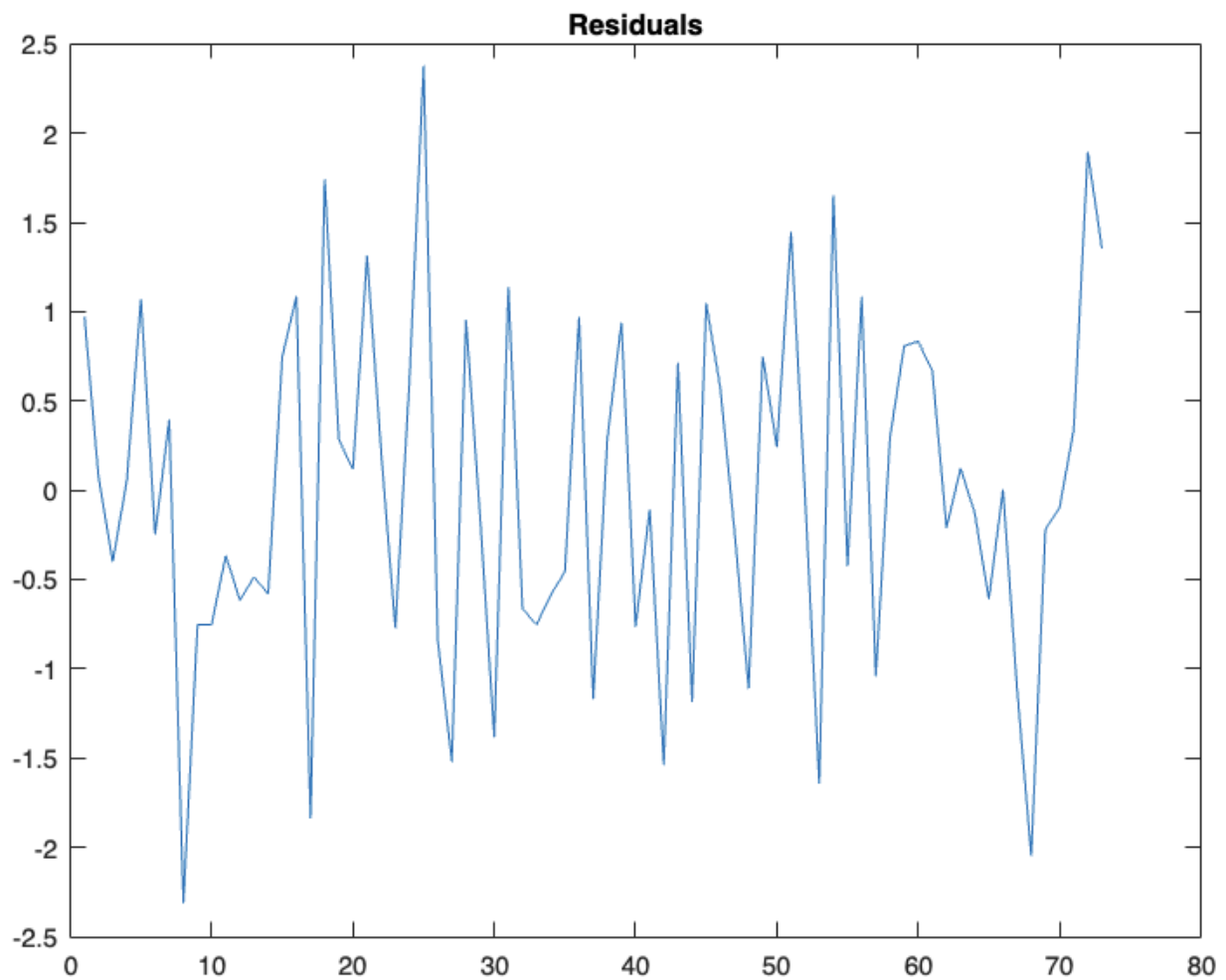
- The **Adjusted R-squared (0.146)** confirms that the overall model fit is still weak even after accounting for model complexity.
- The **RMSE (0.994)** shows that the model's predictions have relatively large errors, indicating the model does not predict the excess return very precisely.

Conclusion:

- **Statistical significance:** The regressor **CEI_134** is not statistically significant, meaning there is not a meaningful relationship between **CEI_134** and **ExcessreturnAsset_134**.
- **Model fit:** The model's fit is **relatively low**, as evidenced by the low R-squared value and high RMSE. While **CEI_134** has some predictive power, the model does not explain a large portion of the variation in excess returns, suggesting that other factors (not included in the model) may be more important in predicting **ExcessreturnAsset_134**.

Residuals

Heteroskedasticity



Interpretation of the Residual Plot:

- If the residuals are randomly scattered around zero with no discernible pattern, this indicates a good fit for the linear regression model.
- If the residuals show a pattern (e.g., trending up or down, or forming a curve), this suggests that the model might be mis-specified or that a non-linear relationship exists between the variables.
- Large residuals (points far from 0) may indicate outliers or data points where the model fails to capture the relationship accurately.

Interpretation:

1. p-value of 0.57779:

Since the p-value is significantly higher than the significance level of 0.01 (typically used to reject the null hypothesis), we do not reject the null hypothesis of homoscedasticity.

This means **there is no evidence to suggest that heteroscedasticity is present** in our model.

2. Breusch-Pagan Test Statistic of 0.30982:

A small test statistic like this suggests that the variance of the residuals **does not depend on the independent variable(s)**. In other words, the residuals' variance is approximately constant across the range of values of the independent variables.

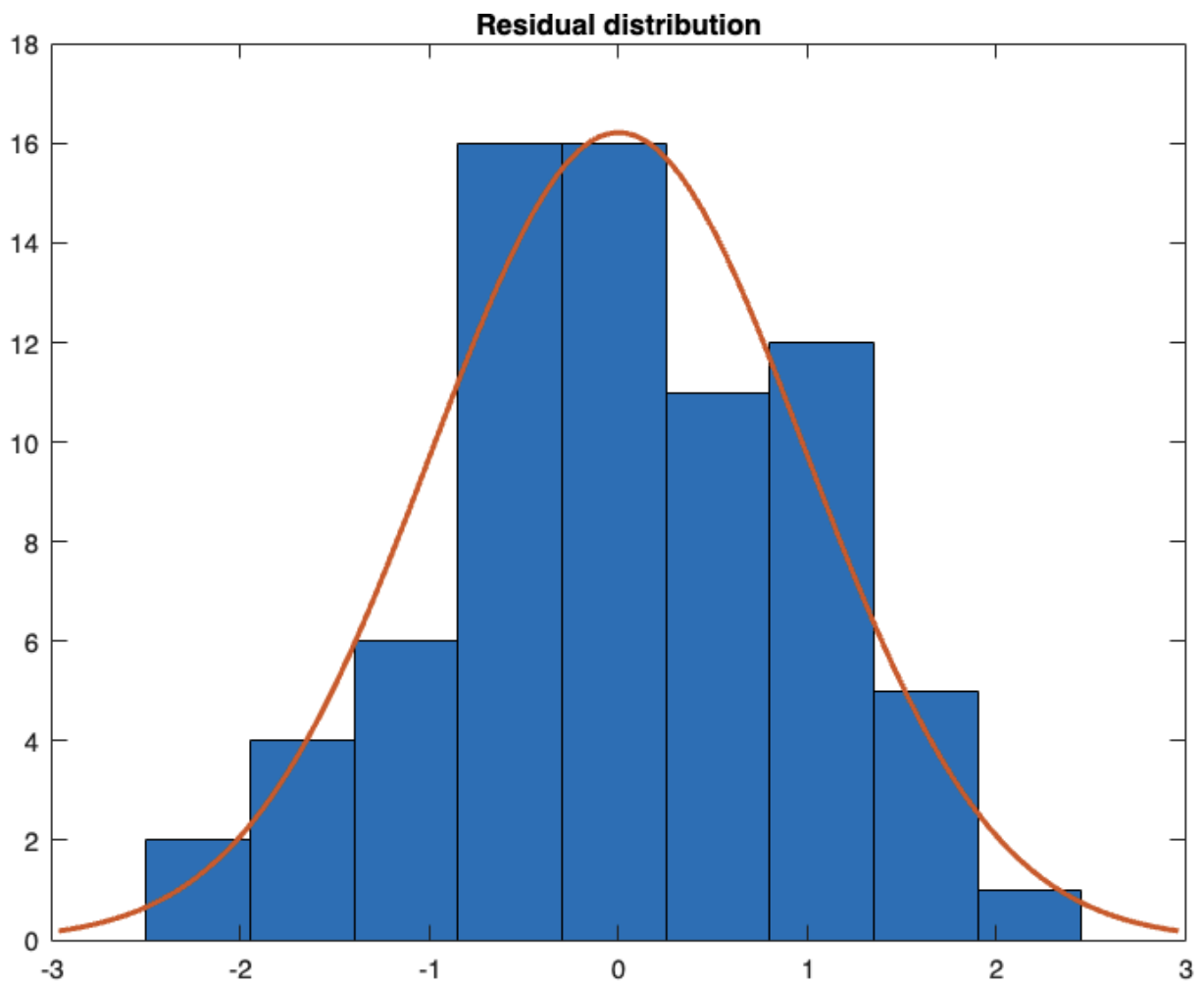
Conclusion:

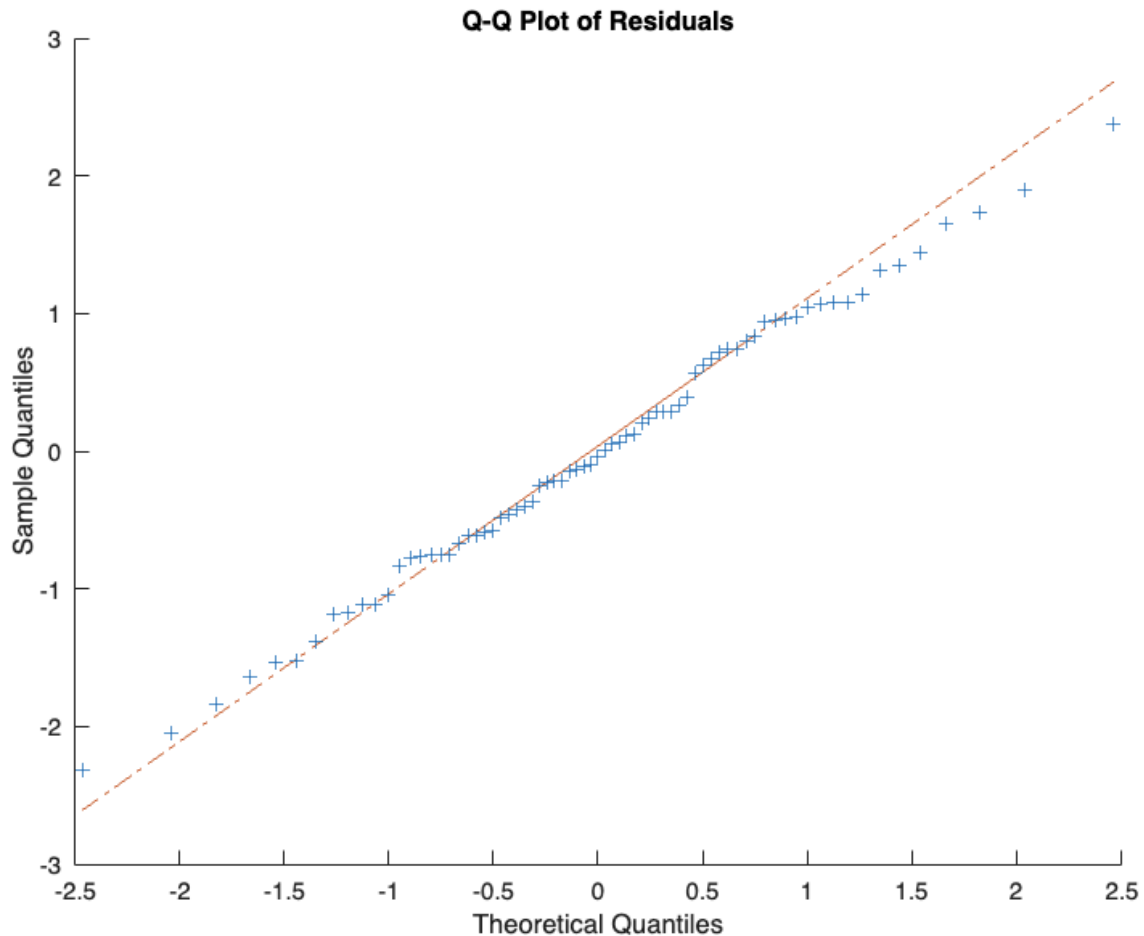
Our model does not show signs of heteroscedasticity, as the p-value is high, and the Breusch-Pagan test statistic is low. This implies that the assumption of homoscedasticity (constant variance of the residuals) holds, which is ideal for the validity of the regression model's results.

In summary:

- The model's residuals are likely homoscedastic, meaning their variance is constant and does not systematically change with the values of the independent variables.
- We can proceed confidently with the results of your regression analysis without worrying about heteroscedasticity affecting your conclusions.

Normality





QQ Plot

The closer the points are to the line, the more they are distributed as Normal, in this case, they are really close to.

JB Test

Interpretation of p-value 0.5 (or even more but Matlab gives warning) in the JB test:

High p-value:

A p-value of 0.5 or higher indicates that the residuals do not deviate significantly from a normal distribution.

The null hypothesis of the JB test is that the residuals are normally distributed. Since the p-value is large, you fail to reject the null hypothesis, meaning the residuals **likely follow a normal distribution**.

K-S Test

Interpretation:

A high p-value (in our case 0.98884) means that there is no evidence to reject the null hypothesis, which assumes that the residuals follow a normal distribution.

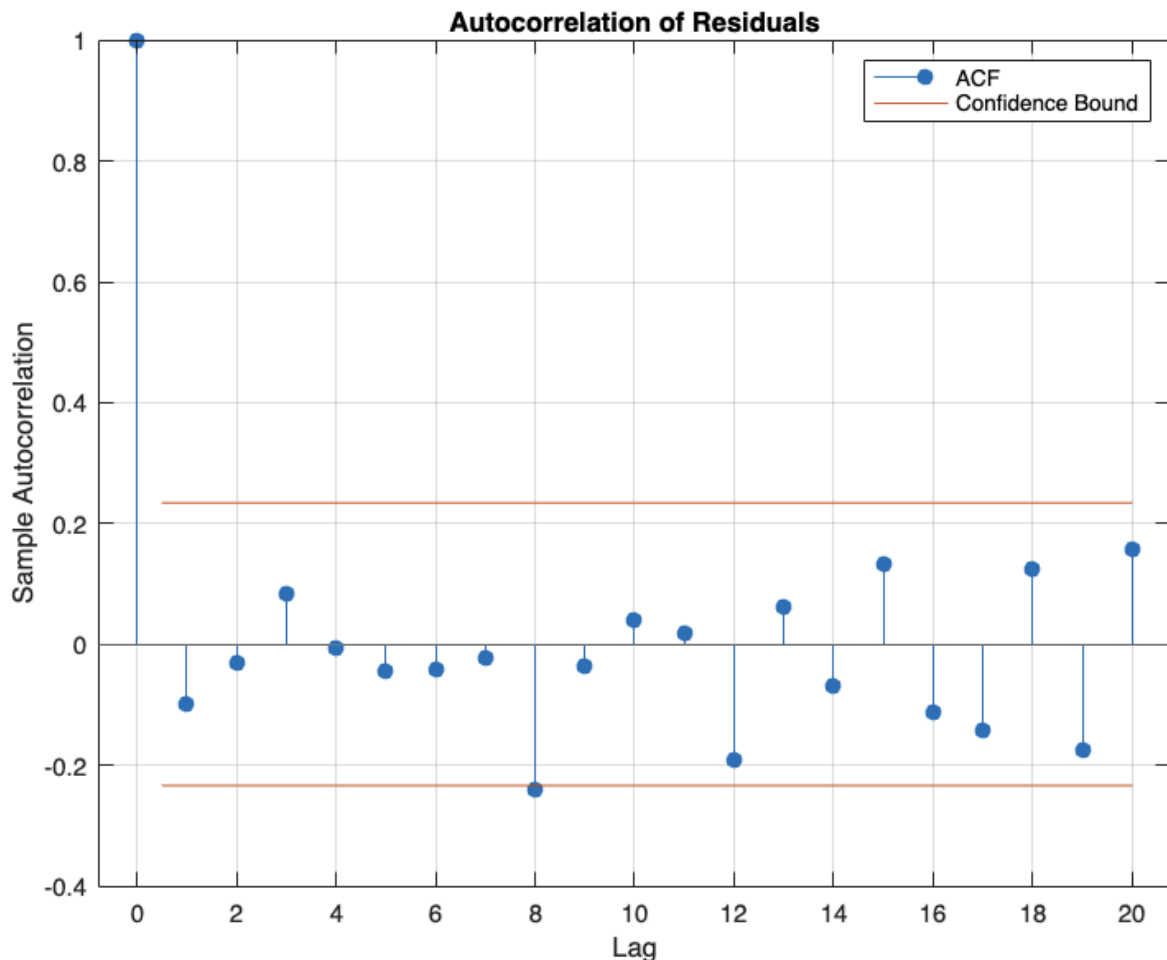
In other words, your data are very likely to follow a normal distribution since the K-S test did not find any significant deviation from normality.

Conclusion:

Our residuals appear to be strongly normal, which is a good sign for our model. It suggests that the assumption of normally distributed residuals holds, making our statistical inferences from the model more reliable.

We can proceed with confidence that the residuals behave as expected under the normality assumption.

Autocorrelation



Low mean autocorrelation (0.0196):

This small value suggests that there is almost no correlation between the residuals at different time lags. In practical terms, the residuals from our model appear to be largely independent of each other.

Independence of residuals:

When residuals are independent (or nearly so), it means that each residual is not influenced by the previous or future values. This is a desirable property in many models, as it indicates that the model has captured most of the structure in the data, leaving only random noise.

Interpretation:

No significant autocorrelation:

The low autocorrelation suggests that there is no persistent pattern in the residuals that would indicate autocorrelated errors. This means that the model does not suffer from **autocorrelation**, which is often a sign that the model is well-specified in capturing the underlying relationships between the variables.

Model validation:

Since our residuals are nearly independent, we can confidently interpret the results of your regression model. The absence of autocorrelation indicates that our model has successfully accounted for temporal relationships, and the **residuals are behaving like white noise**, which is the ideal scenario.

Conclusion:

Our residuals show a low mean autocorrelation (0.0196), meaning they are nearly independent. This is a good sign for model validity, as it suggests that your model is appropriately specified and there is **no indication of autocorrelation issues** in the residuals.

We can proceed with confidence that your model's results and statistical inferences are reliable.

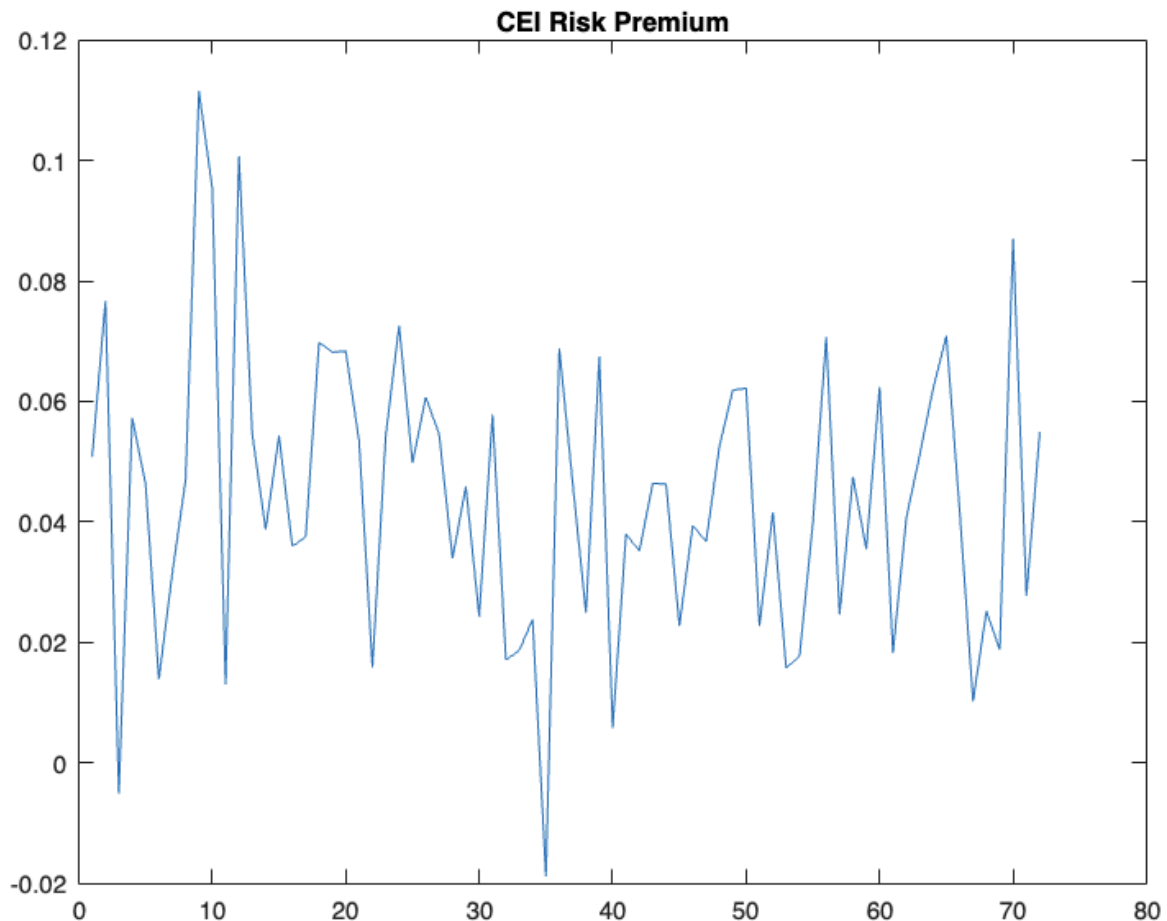
2. Is the CEI risk premium statistically significant?.

Apply the proper test taking into account data characteristics.

Write your answer here

2.

The CEI Risk Premium (CEI_RP)



Is identified as a time series that exhibits stationary characteristics. This stationarity implies that the statistical properties of the series, such as mean and variance, are constant over time, making it suitable for further statistical analysis.

Given these characteristics, we conducted a t-statistic test to evaluate the significance of the CEI_RP. The computed t-statistic value is **15.5563**, which corresponds to a p-value of **0.0000**. This indicates strong evidence against the null hypothesis, suggesting that the CEI_RP is statistically significant.

For our analysis, we have set the significance level (alpha) at **1/73**, which is approximately **0.0137**. Given that our p-value is significantly lower than this alpha level, we can confidently reject the null hypothesis and conclude that the CEI Risk Premium is indeed statistically significant.

Key Findings:

- **It's a stationary time serie**
- **T-statistic:** 15.5563
- **P-value:** 0.0000
- **Alpha Level:** 0.01

3.

Assume that CEI is the only relevant risk factor for AAA stock.

The estimate of the CEI beta of stock AAA is 0.5.

The one-year risk-free rate is 2%.

What is the expected yearly excess return of stock AAA? Explain the result.

Assume that CEI is the only relevant risk factor for BBB stock.

The estimate of the CEI beta of stock BBB is -1.4

The one-year risk-free rate is 2%.

What is the expected yearly excess return of stock BBB?.

Explain the result.

Assume that CEI is the only relevant risk factor for CCC stock.

The estimate of the CEI beta of stock CCC is 1.8

The one-year risk-free rate is 2%.

What is the expected yearly excess return of stock CCC?. Explain the result.

The variance of AAA, BBB and CCC excess returns is 0.23, 0.11 and 0.09, respectively.

The correlations are: (AAA,BBB) -0.34, (AAA,CCC) 0.15, (BBB,CCC) -0.19.

Suggest an investment strategy using AAA, BBB, and CCC stocks based on these results.

Write your answer here

3.

Assets' Performance

	Mean	Max	Min	STD	Skew	Kurt
ExcessReturnAAA	0.0020	0.0358	-0.0294	0.0120	0.1999	3.3538
	Mean	Max	Min	STD	Skew	Kurt
ExcessReturnBBB	-0.0816	0.0064	-0.1762	0.0336	-0.1999	3.3538
	Mean	Max	Min	STD	Skew	Kurt
ExcessReturnCCC	0.0593	0.1808	-0.0540	0.0433	0.1999	3.3538

AAA

- **AAA** has a relatively low mean return (0.2%) and low risk (standard deviation of 1.2%).

- The positive skewness (0.1999) means there are occasional positive outliers in the returns.
- The kurtosis of 3.3538 suggests the return distribution has slightly **heavier tails** than a normal distribution, meaning it may occasionally have extreme returns.

BBB

- **BBB** has a **negative mean return** (-8.16%), meaning it consistently underperforms.
- The standard deviation (3.36%) is higher than **AAA**, indicating more variability in returns.
- Negative skewness (-0.1999) implies **downside risk**—the left tail of the distribution is heavier, indicating more frequent or extreme negative returns.
- Similar kurtosis to AAA, but with a more substantial downside.

CCC

- **CCC** provides the highest mean return (5.93%) but also has the highest standard deviation (4.33%), implying greater **risk**.
- Positive skewness (0.1999) suggests **occasional large positive returns**, and kurtosis (3.3538) implies the presence of occasional extreme returns in both directions, but primarily on the upside.

Given that the **portfolios** were constructed using the **Fama-MacBeth method**, before we can make some conclusions based on the relationship between the betas of each asset and the expected returns:

- **AAA:**

If **AAA** has a **low beta** with respect to the CEI, this implies that **AAA** is less sensitive to changes in carbon emissions and may be more **environmentally resilient**. This could explain why it has a **positive weight** in most portfolios, especially for conservative investors seeking lower risk and exposure to CEI.

- **BBB:**

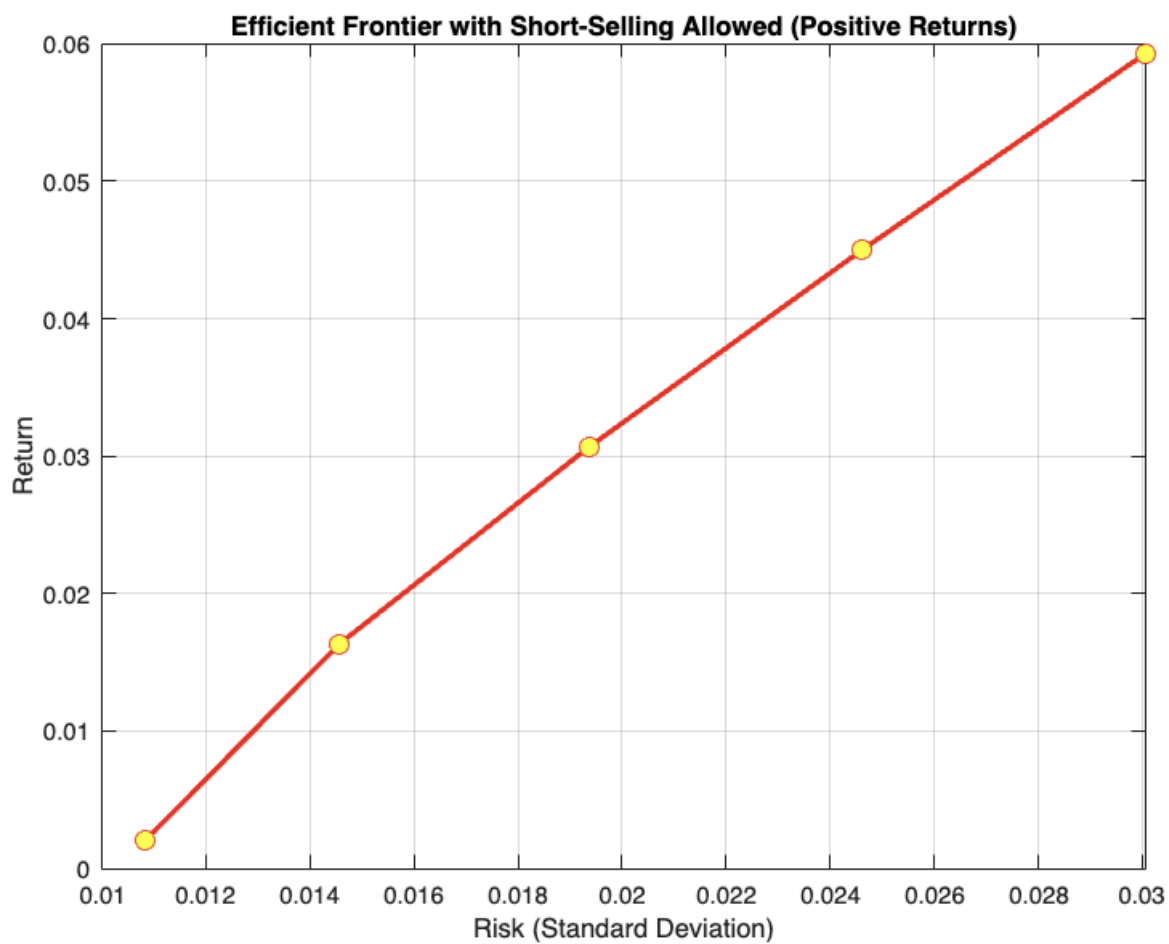
If **BBB** has a **negative beta**, its **negative return** and **negative weight** in most portfolios suggest it might be more **exposed to carbon-intensive industries** or other risks related to CEI. Investors are likely **shorting BBB** to hedge against risks associated with carbon emissions.

- **CCC:**

If **CCC** has a **high beta**, it is more sensitive to **carbon risk**. The fact that **CCC** has **higher returns but higher risk** means that investors expecting high returns may accept its exposure to CEI risks. It is weighted more heavily in the riskier portfolios.

Revised Interpretation of the Portfolios:

With the CEI exposure kept in mind, let's reanalyze the 5 different portfolios we propose:



Portfolio 1:

- Target Return: 0.002
- Risk (Standard Deviation): 0.010827
- Weights: [AAA: 0.82513, BBB: 0.071114, CCC: 0.10375]

Portfolio 2:

- Target Return: 0.016325
- Risk (Standard Deviation): 0.014565
- Weights: [AAA: 0.88209, BBB: -0.053718, CCC: 0.17163]

Portfolio 3:

- Target Return: 0.03065
- Risk (Standard Deviation): 0.019362
- Weights: [AAA: 0.93905, BBB: -0.17855, CCC: 0.2395]

Portfolio 4:

- Target Return: 0.044975
- Risk (Standard Deviation): 0.024606
- Weights: [AAA: 0.99601, BBB: -0.30338, CCC: 0.30737]

Portfolio 5:

- Target Return: 0.0593
- Risk (Standard Deviation): 0.030064
- Weights: [AAA: 1.053, BBB: -0.42821, CCC: 0.37524]

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Portfolio 1 (Low Risk/Low Return):

- **Interpretation:** This portfolio focuses on **low CEI exposure**, favoring AAA, which is less exposed to carbon risk, and has small exposure to **BBB** and **CCC**. It's suitable for risk-averse, environmentally conscious investors.

Portfolio 2 (Moderate Risk/Moderate Return):

- **Interpretation:** This portfolio leans more toward **CCC**, likely due to the higher expected return but still **hedges against carbon risk** by shorting **BBB**, which may have higher CEI exposure.

Portfolio 3 (Balanced Risk/Return):

- **Interpretation:** This portfolio seeks **balanced risk** and return by increasing exposure to **CCC** while further **shorting BBB**. Investors anticipate higher returns while managing CEI risks.

Portfolio 4 (High Risk/High Return):

- **Interpretation:** Investors seek higher returns by increasing **CCC** exposure, reflecting willingness to take on more **CEI risk**, while **shorting BBB** aggressively to mitigate downside risks related to high CEI exposure.

Portfolio 5 (Very High Risk/Very High Return):

- **Interpretation:** This portfolio is the most **CEI-exposed** due to its high weight in **CCC** and shorting of **BBB**, which has a negative return and high exposure to **carbon risk**. Investors here expect high returns, accepting substantial **carbon exposure risk**.

Conclusion:

- **AAA's** consistent positive weighting in all portfolios indicates that it is **low CEI risk**, making it attractive to both conservative and risk-seeking investors.
- **BBB** consistently has **negative weight** due to its **negative return** and high sensitivity to CEI risk, making it a good candidate for shorting.
- **CCC** offers the highest returns but also the highest CEI exposure and risk, making it more suitable for investors with higher risk tolerance who expect to benefit from industries exposed to **carbon risk**.

This strategy aligns with investors' views on the **environmental impact** of carbon emissions, with portfolios offering different levels of **exposure to CEI** based on risk-return preferences.

4. You present your results in a seminar.

A colleague indicates that the CEI premium may be linked to traditional risk factors. She suggests that the CEI risk premium you calculated will be a biased estimate compared to the proper premium.

She points out that data on the most critical risk factor (the market) is available on MarketRP_2024.mat.

Propose a method for finding an appropriate estimate of the CEI premium after adjusting for the market risk factor exposure.

Present the numerical results and discuss them.

Is your colleague right or wrong?

Write your answer here

4.

Correlation Analysis:

Correlation 0.763703490165299 with p value of 6.117937288749957e-15

Interpretation:

The strong positive correlation and significant p-value suggest that the CEI risk premium and the market risk premium are closely related.

Specifically:

- A significant portion of the variation in the CEI risk premium is explained by the market risk premium.
- The large difference between the non-adjusted CEI premium (0.0440) and the adjusted CEI premium (-7.71e-19) is explained by the fact that once market risk is accounted for, there is almost no independent effect of the CEI premium.

Conclusion:

1. CEI and Market Risk are Strongly Correlated: The correlation of 0.7637 shows that the CEI risk premium is heavily influenced by market risk.
2. CEI Premium is Mostly Explained by Market Risk: Given the near-zero adjusted CEI premium after controlling for market risk, most of the CEI premium was driven by market risk.
3. Original CEI Premium Was Likely Biased: Since the adjusted CEI premium is close to zero, the initial estimate of the CEI premium was likely overstated due to its correlation with market risk.

Regression Analysis

Linear regression model:
 $CEI_RP \sim 1 + MarketRP$

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-0.00192853183474702	0.00499552288082097	-0.386052047154288	0.700628997350782
MarketRP	0.556839509238215	0.056259735670627	9.897655980793	6.11793728875088e-15

Number of observations: 72, Error degrees of freedom: 70
Root Mean Squared Error: 0.0156
R-squared: 0.583, Adjusted R-Squared: 0.577
F-statistic vs. constant model: 98, p-value = 6.12e-15

Overall Interpretation of the model:

- **Market risk premium (MarketRP) has a strong and statistically significant impact on the CEI risk premium (CEI_RP). The coefficient for MarketRP (0.5568) is significant, with a very small p-value, indicating that market movements strongly influence the CEI premium.**

- **Intercept is not significant:** The intercept term is not significantly different from zero, meaning that when the market risk premium is zero, the CEI risk premium is also likely to be near zero.
- **Model fit:** The model explains a significant portion of the variation in CEI_RP (R-squared = 0.583), indicating that the market risk premium is a key driver of CEI_RP.

Conclusion:

Our colleague's suggestion that the CEI premium is influenced by market risk appears to be correct. The high correlation and significant coefficient for MarketRP indicate that the CEI risk premium is closely tied to market movements.

This strong link suggests that much of what was initially considered CEI risk is actually driven by market risk.

The next step might be to focus on the adjusted CEI risk premium, which isolates the part of the CEI premium that is independent of market risk.

Interpretation of the adjusted CEI RP:

Adjusted CEI Risk Premium:

The adjusted CEI mean is extremely close to zero ($-7.71e-19$), which suggests that after accounting for market risk, the CEI risk premium has very little to no effect. This implies that any effect the CEI premium initially seemed to have was mostly absorbed by the market risk.

Non-Adjusted CEI Risk Premium:

The initial (non-adjusted) CEI risk premium was 0.0440, which indicates that before considering market risk, the CEI risk premium appeared to have a significant positive effect.

Our answer to colleague intuition

- **Market Risk Explains Most of the CEI Premium:** The strong correlation and significant regression results suggest that most of the variation in the CEI risk premium can be attributed to market risk.
- **Initial CEI Premium Was Biased:** The initial non-adjusted CEI premium was likely overstated because it failed to account for the relationship with market risk.

Extra: the effect on assets' expected return

Asset	CEI Beta	Non-Adjusted CEI RP	Expected Return (Non-Adjusted)	Adjusted CEI RP	Expected Return (Adjusted)
AAA	0.5	0.044	0.022	-7.71e-19	-3.855e-19
BBB	-1.4	0.044	-0.06159999999999999	-7.71e-19	1.0793999999999998e-18
CCC	1.8	0.044	0.07919999999999999	-7.71e-19	-1.3878e-18

If we only considered CEI factor, exp. returns would be around 0 for each asset class, or even negative.

Email to ypenya@eco.uc3m.es your response in a PDF file (Hint: Save a PDF version of your *.mlx file)